Arbitrarily small nonconvex quasihyperbolic balls

A counterexample in a punctured Banach space

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Result. In $X = (\mathbb{R}^2, \|\cdot\|_\infty)$, punctured at the origin, there are nonconvex quasihyperbolic balls of arbitrarily small radius. Thus the general Banach-space critical-radius question in arXiv:1104.3745 has a negative answer.

1 Question and conclusion

Let X be a Banach space and $\Omega = X \setminus \{0\}$. The quasihyperbolic length and distance are

$$\ell_k(\gamma) = \int \frac{\|d\gamma\|}{\|\gamma\|}, \qquad k(x,y) = \inf_{\gamma : x \rightsquigarrow y} \ell_k(\gamma).$$

Klén–Rasila–Talponen ask whether there is, in the Banach-space setting, a positive critical radius below which quasihyperbolic balls in punctured spaces are convex [1, PDF p. 16]. Rasila–Talponen later emphasize that their arbitrarily-small counterexample concerns the distance-ratio metric and that the analogous assertion for quasihyperbolic balls was only presumed [2, PDF p. 11].

2 Proof intuition

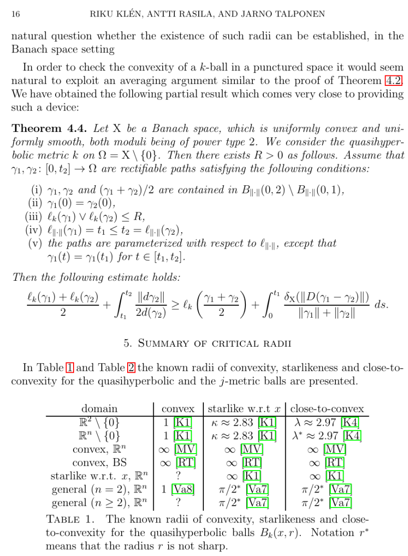
At a fixed face point of the square unit sphere, the most obvious local counterexample fails: the reachable-set boundary is a concave square-root curve, hence the small quasihyperbolic ball is convex. The obstruction instead appears when the center approaches a corner. Two short paths can exploit different dominant coordinates. Their endpoints remain at distance $\frac{3}{2}\delta + \frac{9}{8}\delta^2 + O(\delta^3)$, while every path to their midpoint must cross the diagonal and pays the larger second-order cost $\frac{3}{2}\delta + \frac{81}{64}\delta^2 + O(\delta^3)$.

The lower bound is global. Coordinate sorting is nonexpansive for the infinity norm, so any path can be folded into one which stays on one side of the diagonal before a single crossing and on the other side afterwards. On each side the dominant coordinate controls both coordinate speeds. A sharp one-dimensional envelope estimate then bounds the two pieces of the path.

Theorem 1. *Let $X = (\mathbb{R}^2, \|\cdot\|_\infty)$ and $\Omega = X \setminus \{0\}$. For every $R > 0$ there exist $c \in \Omega$ and $0 < r < R$ such that the quasihyperbolic ball*

$$B_k(c,r) = \{z \in \Omega : k(c,z) < r\}$$

is not convex.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natural question whether the existence of such radii can be established, in the Banach space setting

In order to check the convexity of a k-ball in a punctured space it would seem natural to exploit an averaging argument similar to the proof of Theorem 4.2. We have obtained the following partial result which comes very close to providing such a device:

Theorem 4.4. *Let* X *be a Banach space, which is uniformly convex and uniformly smooth, both moduli being of power type* 2*. We consider the quasihyperbolic metric* k *on* $\Omega = \mathrm{X} \setminus \{0\}$*. Then there exists* $R > 0$ *as follows. Assume that* $\gamma_1, \gamma_2 \colon [0, t_2] \to \Omega$ *are rectifiable paths satisfying the following conditions:*

(i) γ_1, γ_2 *and* $(\gamma_1 + \gamma_2)/2$ *are contained in* $B_{\|\cdot\|}(0,2) \setminus B_{\|\cdot\|}(0,1)$,
(ii) $\gamma_1(0) = \gamma_2(0)$,
(iii) $\ell_k(\gamma_1) \vee \ell_k(\gamma_2) \le R$,
(iv) $\ell_{\|\cdot\|}(\gamma_1) = t_1 \le t_2 = \ell_{\|\cdot\|}(\gamma_2)$,
(v) *the paths are parameterized with respect to* $\ell_{\|\cdot\|}$*, except that* $\gamma_1(t) = \gamma_1(t_1)$ *for* $t \in [t_1, t_2]$.

Then the following estimate holds:

$$\frac{\ell_k(\gamma_1) + \ell_k(\gamma_2)}{2} + \int_{t_1}^{t_2} \frac{\|d\gamma_2\|}{2d(\gamma_2)} \ge \ell_k\left(\frac{\gamma_1 + \gamma_2}{2}\right) + \int_0^{t_1} \frac{\delta_{\mathrm{X}}(\|D(\gamma_1 - \gamma_2)\|)}{\|\gamma_1\| + \|\gamma_2\|}\, ds.$$

5. Summary of critical radii

In Table 1 and Table 2 the known radii of convexity, starlikeness and close-to-convexity for the quasihyperbolic and the j-metric balls are presented.

domain	convex	starlike w.r.t x	close-to-convex
$\mathbb{R}^2 \setminus \{0\}$	1 [K1]	$\kappa \approx 2.83$ [K1]	$\lambda \approx 2.97$ [K4]
$\mathbb{R}^n \setminus \{0\}$	1 [K1]	$\kappa \approx 2.83$ [K1]	$\lambda^* \approx 2.97$ [K4]
convex, $\mathbb{R}^n$	∞ [MV]	∞ [MV]	∞ [MV]
convex, BS	∞ [RT]	∞ [RT]	∞ [RT]
starlike w.r.t. x, $\mathbb{R}^n$	?	∞ [K1]	∞ [K1]
general ($n = 2$), $\mathbb{R}^n$	1 [Va8]	$\pi/2^*$ [Va7]	$\pi/2^*$ [Va7]
general ($n \ge 2$), $\mathbb{R}^n$	?	$\pi/2^*$ [Va7]	$\pi/2^*$ [Va7]

Table 1. The known radii of convexity, starlikeness and close-to-convexity for the quasihyperbolic balls $B_k(x, r)$. Notation r^* means that the radius r is not sharp.

Figure 1: The Banach-space convexity-radius discussion in arXiv:1104.3745, PDF page 16.

3 Two elementary path estimates

We use the following sharp envelope bound.

Lemma 1 (exponential tent bound)**.** *Suppose* $u : [0, T] \to (0, \infty)$ *is absolutely continuous,* $|u'| \le u$ *almost everywhere,* $u(0) = a$*, and* $u(T) = b$*. Then*

$$u(t) \le \min\{ae^t, be^{T-t}\}$$

and consequently

$$\int_0^T u(t)\, dt \le 2e^{T/2}\sqrt{ab} - a - b.$$

Proof. The function $\log u$ is 1-Lipschitz, giving the two pointwise bounds. Their graphs meet at $t_0 = (T + \log(b/a))/2$, which lies in $[0, T]$ because necessarily $T \ge |\log(b/a)|$. Integrating the two exponentials on the corresponding subintervals gives the displayed formula. ∎

We also use the standard radial lower bound

$$k(x, y) \ge \left| \log \frac{\|y\|_\infty}{\|x\|_\infty} \right|. \tag{1}$$

Indeed, the norm is absolutely continuous along a rectifiable path and the absolute derivative of its logarithm is bounded by the quasihyperbolic speed.

4 The explicit configuration

Fix a sufficiently small $\delta > 0$ and put

$$c_\delta = (1, 1-\delta), \qquad q_0 = 1 - \frac{\delta}{2},$$

$$A_\delta = \left(1 - \frac{3\delta}{2}, 1 - 2\delta\right), \qquad B_\delta = \left(1 - \frac{3\delta}{2} - \frac{3\delta^2}{4}, 1 - \frac{\delta}{2}\right),$$

$$M_\delta = \frac{A_\delta + B_\delta}{2} = \left(1 - \frac{3\delta}{2} - \frac{3\delta^2}{8}, 1 - \frac{5\delta}{4}\right).$$

Thus M_δ is the midpoint of the two proposed in-ball points. Define

$$L_A(\delta) = -\log\left(1 - \frac{3\delta}{2}\right), \tag{2}$$

$$g_B(\delta) = (B_\delta)_2 - (B_\delta)_1 = \delta + \frac{3\delta^2}{4},$$

$$L_B(\delta) = -\log q_0 + 2\log\left(1 + \frac{g_B(\delta)}{2q_0}\right), \tag{3}$$

$$g_M(\delta) = (M_\delta)_2 - (M_\delta)_1 = \frac{\delta}{4} + \frac{3\delta^2}{8},$$

$$L_M(\delta) = -\log q_0 + 2\log\left(\frac{2q_0 + g_M(\delta)}{2\sqrt{q_0 (M_\delta)_2}}\right). \tag{4}$$

4.1 Upper bound for A_δ

Let $T = L_A(\delta)$ and, for $0 \le t \le T$, set

$$x(t) = e^{-t}, \qquad y(t) = 1 - \delta - \frac{2}{3}(1 - e^{-t}).$$

This path joins c_δ to A_δ. Moreover $x(t) > y(t) > 0$ for small δ, $\max\{|x'|, |y'|\} = x$, and $\|(x,y)\|_\infty = x$. Its quasihyperbolic speed is therefore one, so

$$k(c_\delta, A_\delta) \le L_A(\delta). \tag{5}$$

4.2 Upper bound for B_δ

First follow

$$x(t) = e^{-t}, \qquad y(t) = 2 - \delta - e^{-t}, \qquad 0 \le t \le -\log q_0.$$

This has unit quasihyperbolic speed, stays in $x \ge y$, and ends at (q_0, q_0). Set

$$T_B = 2\log\left(1 + \frac{g_B(\delta)}{2q_0}\right)$$

and define, for $0 \le t \le T_B$,

$$Y(t) = q_0 e^{\min\{t, T_B - t\}}, \qquad X(t) = q_0 - \int_0^t Y(s)\, ds.$$

Here $Y \geq q_0 \geq X$, and $\max\{|X'|, |Y'|\} = Y = \|(X,Y)\|_\infty$ almost everywhere. Also

$$\int_0^{T_B} Y(s)\, ds = 2q_0(e^{T_B/2} - 1) = g_B(\delta),$$

so the second path ends at B_δ. Concatenation yields

$$k(c_\delta, B_\delta) \leq L_B(\delta). \tag{6}$$

5 A path-independent lower bound for the midpoint

We prove

$$k(c_\delta, M_\delta) \geq L_M(\delta). \tag{7}$$

For small δ, $L_M(\delta) < 1/4$. A path of quasihyperbolic length less than $1/4$ starting at c_δ remains in the positive quadrant: after quasihyperbolic arclength t, its norm is at most e^t, hence its ordinary displacement is at most $te^t < 1/2$. Longer paths already satisfy (7).

Consider therefore a path in the positive quadrant from c_δ, where $x > y$, to M_δ, where $y > x$. Choose a time at which it meets the diagonal. Before that time apply the coordinate-sorting map $(x, y) \mapsto (\max\{x, y\}, \min\{x, y\})$; afterwards apply the reverse sorting map. These maps fix the diagonal, preserve the infinity norm, and are 1-Lipschitz. Thus the folded path has no greater quasihyperbolic length, stays in $x \geq y$ before one crossing (q, q), and stays in $y \geq x$ afterwards.

Let its two lengths be T_1, T_2. On the first piece the dominant coordinate is x, so both coordinate speeds are bounded by x. Lemma 1 and the net increase of the second coordinate give

$$q - (1 - \delta) \leq 2e^{T_1/2}\sqrt{q} - 1 - q.$$

Consequently, writing $q_0 = 1 - \delta/2$,

$$T_1 \geq \begin{cases} -\log q, & q \leq q_0, \\ 2\log \dfrac{2q + \delta}{2\sqrt{q}}, & q \geq q_0. \end{cases} \tag{8}$$

The first line is (1); the second is the preceding envelope inequality.

Put $y_M = (M_\delta)_2$ and $g = g_M(\delta) = y_M - (M_\delta)_1$. On the second piece the dominant coordinate is y. The required decrease of the first coordinate and Lemma 1 imply

$$q - (M_\delta)_1 \leq 2e^{T_2/2}\sqrt{q y_M} - q - y_M,$$

hence

$$T_2 \geq 2\log \frac{2q + g}{2\sqrt{q y_M}}. \tag{9}$$

If $q \leq q_0$, adding (8) and (9) gives

$$T_1 + T_2 \geq -\log y_M + 2\log\left(1 + \frac{g}{2q}\right),$$

which decreases as q increases and is therefore minimized at q_0. If $q \geq q_0$, the lower bound is

$$2\log \frac{(2q + \delta)(2q + g)}{4q\sqrt{y_M}}.$$

Apart from constants independent of q, its argument is

$$\frac{(2q+\delta)(2q+g)}{q} = 4q + 2(\delta+g) + \frac{\delta g}{q}.$$

Its derivative is $4 - \delta g/q^2 > 0$ for all $q \geq q_0$ when δ is small. This branch is likewise minimized at q_0. At $q = q_0$ the common lower bound is exactly (4), proving (7).

6 The second-order separation

Taylor expansion at $\delta = 0$ gives

$$L_A(\delta) = \frac{3}{2}\delta + \frac{9}{8}\delta^2 + O(\delta^3),$$
$$L_B(\delta) = \frac{3}{2}\delta + \frac{9}{8}\delta^2 + O(\delta^3),$$
$$L_M(\delta) = \frac{3}{2}\delta + \frac{81}{64}\delta^2 + O(\delta^3).$$

Since $81/64 - 9/8 = 9/64 > 0$, for every sufficiently small positive δ,

$$\max\{L_A(\delta), L_B(\delta)\} < L_M(\delta).$$

Choose

$$r_\delta = \frac{max\{L_A(\delta), L_B(\delta)\} + L_M(\delta)}{2}.$$

Equations (5), (6), and (7) show that $A_\delta, B_\delta \in B_k(c_\delta, r_\delta)$ but their midpoint M_δ is not in the ball. Finally $r_\delta \to 0$, proving Theorem 1.

7 Scope and literature check

The counterexample answers the unrestricted Banach-space question. It does not contradict positive results under Hilbertian, uniformly convex, or uniformly smooth hypotheses: the infinity norm is deliberately nonsmooth and not strictly convex. The related paper [2] proves an averaging estimate under power-type-2 assumptions but explicitly says it does not obtain a critical radius for punctured Banach spaces.

A bounded search through 11 August 2026 used the exact question wording, "critical radius," "punctured Banach space," and quasihyperbolic convexity keywords, together with the citation neighborhood of arXiv:1007.3197 and arXiv:1104.3745. It found Luiro's later solution of the Euclidean finite-dimensional convexity conjecture [3], but no later answer to the Banach-space question treated here. The present result is therefore recorded as an agent counterexample, subject to expert verification and normal novelty review.

References

[1] R. Klén, A. Rasila, and J. Talponen, *Quasihyperbolic Geometry in Euclidean and Banach Spaces*, arXiv:1104.3745 (2011), `https://arxiv.org/abs/1104.3745`.

[2] A. Rasila and J. Talponen, *Convexity properties of quasihyperbolic balls on Banach spaces*, Ann. Acad. Sci. Fenn. Math. 37 (2012), 215–228, arXiv:1007.3197, `https://arxiv.org/abs/1007.3197`.

[3] H. Luiro, *On the uniqueness of quasihyperbolic geodesics*, arXiv:1504.01981 (2015), https://arxiv.org/abs/1504.01981.